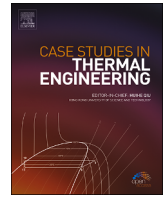




Contents lists available at ScienceDirect

Case Studies in Thermal Engineering

journal homepage: www.elsevier.com/locate/csie



Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions

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ARTICLE INFO

Handling Editor: Huihe Qiu

Keywords:

PV panel
Machine learning
Output power
SVMR
MR
GR

ABSTRACT

Solar photovoltaic (PV) panels play a crucial role in sustainable energy generation, yet their power output often faces uncertainties due to dynamic weather conditions. In this study, a comparative machine learning approach is introduced, utilizing multivariate regression (MR), support vector machine regression (SVMR), and Gaussian regression (GR) techniques for precise solar PV panel power prediction. The investigation into the impact of environmental factors—solar radiation, ambient temperature, and relative humidity—on PV panel output reveals the superior predictive capabilities of SVMR models. With a mean squared error (MSE) of 0.038, a mean absolute error (MAE) of 0.17, and an R^2 value of 0.99, SVMR outperforms GR and MR models. Conversely, Gaussian regression demonstrates comparatively weaker performance, yielding an R^2 of 0.88, an MSE of 0.49, and an MAE of 0.63.

This research underscores the reliability and enhanced accuracy of the proposed SVMR model in forecasting solar PV panel output. The outcomes presented herein carry significant implications for promoting the widespread adoption of PV panels in electricity generation, particularly in challenging environmental conditions. The findings offer valuable insights into optimizing solar PV deployment, ultimately contributing to the expansion of solar power generation in the national energy landscape. Moreover, the comparative analysis provides insights into how anti-

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<https://doi.org/10.1016/j.csie.2024.104459>

Received 28 January 2024; Received in revised form 16 April 2024; Accepted 26 April 2024

Available online 1 May 2024

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pated PV power generation can adapt to varying weather conditions, encompassing factors such as temperature, humidity, and solar radiation.

Abbreviations	
PV	Photovoltaic
ML	Machine Learning
SVMR	Support Vector Machine Regression
MR	Multivariate Regression
GPR	Gaussian Process Regression
MSE	Mean Squared Error
MAE	Mean Absolute Error
NOCT	Normal Operating Cell Temperature

1. Introduction

The escalating consumption of non-renewable fossil fuels underscores the critical need to conserve electrical energy, given their significant contribution to global warming and climatic issues. Consequently, the search for sustainable and clean sources of electricity generation has become paramount, with solar-based power emerging as a promising contender [1–3]. Solar energy's appeal lies in its abundant availability and eco-friendly nature, making it a viable alternative to traditional fossil fuels. The progressive reduction in the cost of large photovoltaic (PV) panels has further facilitated their widespread adoption across numerous regions [4,5].

To promote solar energy utilization, the Indian government has implemented various support measures, with rooftop-based PV power plants playing a crucial role in enhancing sustainable energy production [6,7]. However, transitioning to solar power presents unique challenges, notably the variability of weather conditions impacting PV panel output. The uncertainty surrounding PV panel systems' power output poses economic hurdles, emphasizing the paramount importance of accurate prediction models to support efficient economic models for PV panel system operation [8,9]. Meteorological conditions, including cloud cover, solar radiation, air temperature, humidity, and dust accumulation, significantly influence PV panel performance [10,11].

Addressing these challenges involves implementing various strategies. Thermal management techniques have been devised to optimize PV panel system efficiency, aiming to mitigate issues related to temperature fluctuations [12–14]. Additionally, dust accumulation poses a significant obstacle by blocking solar radiation, thereby reducing panel performance. Moreover, high humidity levels have a detrimental effect on PV panel output power, leading to decreased performance [15]. Furthermore, as multiple PV systems are integrated into hybrid energy solutions, there's a critical need for effective switching technology. This technology ensures seamless transition between available energy sources, highlighting the importance of accurate prediction models for PV panel output power [16,17].

In recent years, machine learning (ML) approaches have gained prominence in predicting PV panel performance. These ML models provide accurate prediction results within shorter timescales, further enhancing the efficiency and reliability of solar energy systems [18,19]. Despite these advancements, the current state-of-the-art in PV power output prediction remains a topic of ongoing research. Various ML methods/models have been proposed, but their design and effectiveness in real-world applications require thorough assessment. This research undertakes a comparative exploration of distinct ML methods to enhance PV power output prediction, with a focus on developing an ML-based prediction model tailored for rooftop solar power plants. By assessing the current state-of-the-art and evaluating the effectiveness of different ML techniques, this study aims to provide valuable insights into the best practices for PV power output prediction under varying environmental conditions.

2. Advances in photovoltaic panel power prediction: a comprehensive review

The anticipation of photovoltaic (PV) solar module output power has witnessed significant exploration in recent years, predominantly employing diverse machine learning methodologies. Among the notable advancements, hybrid models have garnered attention for their potential in enhancing prediction accuracy. In certain studies [20,21], the integration of three prediction modules merits discussion, wherein two models cater to numerical forecasts and one capitalizes on artificial intelligence for PV panel predictions. However, the intricate nature and resource-intensive characteristics of such models may compromise their user-friendliness [22,23].

Prediction strategies in this context may be broadly classified into deterministic and non-deterministic algorithms. In the deterministic paradigm, the prediction algorithm consistently produces identical outputs for a given input, adhering to a predetermined sequence of states. Conversely, non-deterministic approaches consider variations in the environment when predicting outcomes, potentially yielding diverse results even with identical input data [24]. Notably, the operation of PV solar modules tends to display non-deterministic behavior. Performance depends not only on various atmospheric parameters but is also affected by fluctuations in climatic conditions. The same set of atmospheric parameters may result in different outcomes in cloudy and sunny climates, challenging the effectiveness of many deterministic prediction approaches in estimating PV solar module performance [25].

Consequently, the development of an accurate prediction model for specific PV panel installations necessitates the consideration of numerous input parameters. Variables such as date, month, time, geographical location, ambient temperature, humidity, and solar radiation are integral in this endeavor. Previous studies have predominantly relied on deterministic approaches for predicting solar

panel output power. Nevertheless, the inherent uncertainty within power data often curtails the attainment of satisfactory prediction results [26,27].

Despite the limited volume of research on PV power prediction, coupled with the variability of environmental conditions, ample opportunities remain to enhance prediction accuracy through the exploration of diverse machine learning approaches. Recent progress in applying machine learning techniques to PV performance prediction has demonstrated promise. Consequently, several studies have probed potential challenges and avenues for further research [28,29]. A body of contemporary work has contributed a diversity of approaches for PV power prediction and estimation, reflecting the evolving landscape of this research domain. For a comprehensive overview of literature on the estimation of PV system performance, please refer to Table 1.

The selection of Support Vector Machine Regression (SVMR), Gaussian Process Regression, and Multivariate Regression was driven by several considerations. Firstly, SVMR is renowned for its robust performance in handling complex datasets and nonlinear relationships, making it well-suited for capturing the intricate interplay of environmental factors affecting PV panel performance. Its ability to effectively model nonlinear relationships aligns with the dynamic and multifaceted nature of solar energy generation.

Gaussian Process Regression offers a probabilistic framework for regression tasks, allowing for uncertainty estimation in predictions. This is particularly valuable in the context of PV power prediction, where uncertainties stemming from environmental variability can significantly impact model accuracy. By incorporating Gaussian Process Regression, we aim to develop prediction models that not only provide accurate point estimates but also quantify prediction uncertainty, enabling more informed decision-making. Additionally, Multivariate Regression allows for the simultaneous consideration of multiple predictors, facilitating the integration of various environmental parameters such as solar radiation, temperature, humidity, and geographical location. By leveraging Multivariate Regression, we aim to develop a comprehensive predictive model that accounts for the combined effects of these diverse factors, thereby enhancing the accuracy and reliability of our predictions.

Furthermore, these techniques are well-established with strong theoretical foundations and extensive empirical validation in diverse domains, including solar energy prediction. Their versatility and computational efficiency make them practical for real-time applications in the context of solar energy management.

3. Experiments and data collection

The experiments were carried out at the premises of Aditya Engineering College's KLR Bhavan, situated at coordinates 17.05°N and 82.06°E within the Indian state of Andhra Pradesh. The investigation centered on a solar installation featuring two distinct solar stations, each incorporating a 3.2-kW (kW) solar photovoltaic (PV) system. Within the configuration of a 3.2 kW system, each station comprises four rows, and each row houses 24 PV panels, with a power rating of 325 W each (ELDORA VSP 723250304 model), linked in series. Technical specifications for the PV panels are provided in Table 2 for reference. Notably, the study focused on a single 325 W PV panel to prevent any interference with the operation of the remaining panels, as measurements of the panel's DC output power were obtained.

The visual representation of the solar station's layout can be observed in Fig. 1. The research protocol involved the collection of data over a span of 60 days, specifically during the months of January, February, and March in 2022. During this period, key environmental parameters, namely ambient temperature, relative humidity, solar radiation, and output power, were recorded at the precise

Table 1
Past research work on the prediction of the output power of solar PV panels.

Sl. No	References	Year	Type of Work
1	[30]	2018	Prediction of PV power using machine learning approach
2	[31]	2019	Prediction of PV Panel power under the influence of different environmental parameters using ML
3	[32]	2020	Development of software tool to predict and manage the solar power produced by solar panels
4	[33]	2021	Classification of PV panels using deep neural models
5	[34]	2022	Presentation of solar backup system to provide reliable energy using an intelligent prediction-based controller
6	[35]	2022	Solar power forecast based on data models and performance metrics
7	[36]	2022	Prediction of PV power prediction under snow conditions using systematic analysis and computational intelligence modelling
8	[37]	2022	Diagnosis of dust deposition for PV panel surface using the multi-scale-ResNet approach
9	[38]	2022	Forecasting of the output power of the panel using a Tree-based machine learning algorithm
10	[39]	2022	Development of topology mechanism for optimizing the PV panel systems using machine learning

Table 2
Specification of 325 W solar PV panel.

Sl. No	Parameter	Value	Unit
1	Rated peak power	325	Watt
2	Open circuit voltage	46.2	Volt
3	Short circuit current	9.13	Amp
4	Rated voltage	37.8	Volt
5	Rated current	8.6	Amp
6	Fill factor	77.07	%
7	Fire rating	Type	Class C
8	NOCT	44 ± 2	°C



Fig. 1. Photo view of solar power station.

hour of 12 noon. The selection of these months was strategic, aimed at minimizing the influence of rainfall, which, notably, was absent during this timeframe at the chosen site. The selection of 12 noon as the designated data collection time was based on a preliminary seven-day pilot study. This preliminary study involved hourly measurements of horizontal solar radiation falling upon the panel surface, conducted from 9 a.m. to 5 p.m. Solar radiation measurements were executed using a TM-207 solar power meter, a device designed for quantifying direct solar radiation incident on a horizontal surface. A graphical representation of solar radiation with respect to time can be discerned in Fig. 2, illustrating the peak solar radiation values occurring around 12 noon. Thus, the 12 noon time-frame was adopted for measuring the output power of the PV panels across various environmental conditions.

Each day throughout the study period involved the recording of atmospheric parameters, encompassing ambient temperature, relative humidity, and solar radiation. Concurrently, the output power of the PV panels was computed based on output voltage and current values. Given the inherent variability in cloud cover, atmospheric parameters exhibited fluctuations, consequently influencing daily power output. The comprehensive field study spanning 60 days sought to enhance the accuracy of the prediction model by accounting for the effects of these atmospheric variations. The decision to incorporate specific environmental parameters in the model was driven by multiple considerations. Firstly, the authors aimed to construct a predictive model capable of capturing the most influential variables affecting PV module performance while ensuring simplicity and computational efficiency. Secondly, parameter selection was influenced by the availability and reliability of data sources suitable for real-time forecasting applications.

While parameters like pressure can impact PV module efficiency over the long term, their direct effect on short-term forecasting accuracy may be comparatively minor in relation to factors such as solar irradiance and ambient temperature. Additionally, the inclusion of additional parameters could introduce complexity and computational overhead without commensurate improvements in forecasting accuracy within the study's scope.

4. Machine learning model

The schematic representation of the proposed machine learning model for predicting PV panel performance is illustrated in Fig. 3. The model's development commences with the collection of a dataset through meticulously conducted experiments. This dataset encompasses diurnal performance data, capturing the temporal variations in power generation. Key independent variables comprising ambient temperature, humidity, and solar radiation are integral to this dataset, enabling a comprehensive analysis of performance determinants. Once the dataset is assembled, a critical data preprocessing stage ensues. Outliers and zero values are systematically eliminated to ensure the dataset's integrity. To enhance prediction accuracy, feature engineering is employed to identify and select the

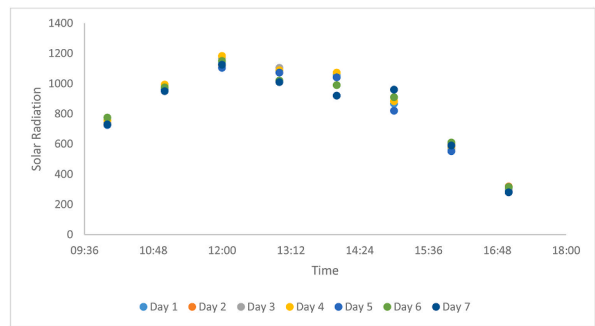


Fig. 2. Variation of solar radiation with time in a day.

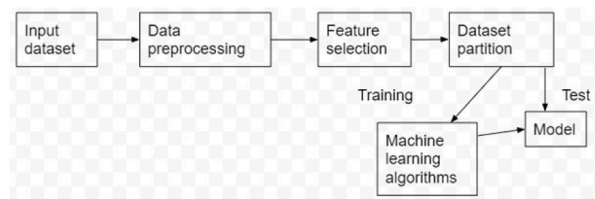


Fig. 3. Flowchart of the proposed work.

most pivotal feature vectors from the acquired dataset. These chosen feature vectors serve as the foundation for predicting performance, a crucial step in the model's development.

Subsequently, the dataset is partitioned into two distinct segments: an 80 % training subset and a 20 % testing subset. This division is fundamental for the rigorous evaluation of the model's predictive capabilities. Three distinct regression algorithms are enlisted for performance prediction: multivariate regression, Support Vector Regression (SVR), and Gaussian regression. Each of these models is rigorously trained using the training dataset, enabling them to learn and adapt to the patterns within the data. Following the comprehensive training phase, the performance of each model is systematically evaluated. This assessment involves the utilization of the reserved 20 % testing dataset, enabling the models to demonstrate their predictive prowess. Performance metrics are systematically computed, delivering a quantitative gauge of each model's accuracy and reliability. The forthcoming sections delve into an in-depth description of each regression algorithm, offering a comprehensive insight into their methodologies and contributions to the prediction model.

4.1. Multivariate regression

Multivariate regression represents a vital extension of simple linear regression, offering an advanced framework for predicting the value of a dependent variable based on the influence of multiple independent variables. This method is particularly pertinent in scenarios where the outcome of interest is influenced by a combination of factors, each contributing a distinct degree of impact. The foundational mathematical expression for multivariate regression is encapsulated in Equation (i) [40].

$$Y = \beta_0 + \beta_1 \beta_n X_n \quad (i)$$

Where $\beta_0, \beta_1, \dots$ are the coefficients and $X_1, X_2, \dots$ are the independent variables. Central to the efficacy of multivariate regression is the quantification of its predictive accuracy. This assessment is achieved through the calculation of the Mean Squared Error (MSE), a pivotal cost function characterizing the model's predictive performance. The formulation of the MSE, as illustrated in Equation (ii), encapsulates the essence of model evaluation.

$$MSE = \frac{1}{2m} \sum_{i=1}^m (Y_i^p - Y_i)^2 \quad (ii)$$

Where Y_i^p is the predicted value of the dependent variable; Y_i is the actual value of the dependent variable.

4.2. Support vector machine regression

Support Vector Machine Regression (SVMR) stands as a supervised machine learning algorithm capable of performing both classification and regression tasks with precision. While its classification counterpart utilizes SVM for distinct categorization, SVMR specifically addresses regression problems. This section provides a comprehensive technical overview of SVMR, elucidating its fundamental components and mathematical formulations.

At the core of SVMR is the concept of the hyperplane, which serves as the fitting line for input data, enabling the algorithm to predict the values of the target or dependent variable. SVMR involves several critical parameters, including the hyperplane, kernel function, boundary lines, and support vectors. The hyperplane plays a pivotal role in SVMR, allowing for the prediction of the dependent variable. The kernel function, on the other hand, is responsible for transforming lower-dimensional data into higher-dimensional space. This dimensional expansion enhances the model's capacity to capture complex relationships within the data, thereby improving predictive accuracy.

In SVMR, data points are positioned within the context of boundary lines. In the case of SVMR with lower dimensions, two boundary lines are employed, collectively forming a margin. The primary objective in SVMR is to minimize the distance between these boundary lines, a critical task for enhancing predictive precision. This optimization process ensures that the predicted values align closely with the actual observations. Central to SVMR are the "support vectors," a subset of data points located in close proximity to the hyperplane. These support vectors play a pivotal role in shaping the position and orientation of the hyperplane, significantly influencing the accuracy of the regression model. The essential mathematical foundations of SVMR are encapsulated in equation (iii) and equation (iv), defining the model's objective function and constraints, respectively. These equations represent the core of SVMR, outlining the intricate optimization process central to its operation [41,42].

$$\frac{1}{2} \|w\|^2 \quad (iii)$$

$$|y_i - w_i x_i| \ll \epsilon \quad (iv)$$

Equation 3 quantifies the objective function in SVMR, aiming to minimize it, while Equation 4 defines the constraints, setting a permissible error margin ϵ for the deviations between predicted and actual values. The process aims to achieve the optimal placement of the hyperplane and margin, ensuring accurate predictions and maximizing the regression model's effectiveness.

4.3. Gaussian Process Regression

It is a nonparametric Bayesian approach to predicting the value of a dependent variable. The GPR technique having a number of key features such as it performs well on small datasets and it also provides uncertain measurements on the prediction of dependent

variable value. The learning of Bayesian approach is based on probability distribution over all the possible values. The Bayesian approach prior distribution is defined as in equation (5) [43].

$$p(w|y, x) = \frac{p(y|X, w)p(w)}{p(y|x)} \quad (v)$$

Where, $p(w|y, x)$ is called posterior distribution.

$p(X, w)$ is called likelihood; $p(w)$ is called prior; $p(x)$ is called marginal likelihood

Fig. 4 provides a comprehensive flowchart depicting the correlation among all the feature vectors integral to our analysis. The dataset under consideration comprises four independent feature vectors: day, ambient temperature, humidity, solar radiation, and the dependent feature vector, power. This results in a total of 25 correlations among these feature vectors, as visually presented in Fig. 4. Of particular significance, Fig. 4 prominently highlights a positive correlation between solar radiation and power consumption. This observation underscores the critical role of solar radiation as a key feature vector within the dataset. Solar radiation's influence on power consumption is evident, making it a central component in the predictive model. Conversely, the other feature vectors exhibit notably weaker correlations when considered in pairs, reaffirming the singular importance of solar radiation in predicting power values. As such, the predictive power of solar radiation emerges as a cornerstone within our dataset, guiding our regression analysis.

5. Results and discussion

In this section, we delve into the outcomes of our study and engage in a comprehensive discussion of the performance of distinct machine learning algorithms concerning solar power prediction. The study commenced with the meticulous collection of a dataset, a pivotal step in our investigation. To assess the performance of solar panels, a multitude of measurements were conducted under diverse environmental conditions.

5.1. Dataset overview

The study commenced with the meticulous collection of a comprehensive dataset, marking a crucial step in the evaluation of machine learning algorithms for solar power prediction. The dataset encompasses a range of independent variables, including ambient temperature, humidity, solar radiation, panel temperature, voltage, and current. Together with the dependent variable of power, these variables establish the basis for assessing algorithm performance. Descriptive statistics of the dataset are presented in Table 3, offering a summary of key attributes such as ambient temperature, humidity, solar radiation, power output, PV panel temperature, voltage, and current. These statistics provide insights into the central tendencies and variability of each variable, thereby contributing to a comprehensive understanding of the dataset within the context of the analysis. Table 4 shows that correlation among the different feature vectors.

5.2. Data preprocessing and exploration

The research proceeded through a tripartite process encompassing data refinement, exploratory analysis, and correlation examination. Initially, rigorous attention was directed towards refining the dataset by removing feature vectors characterized by constant

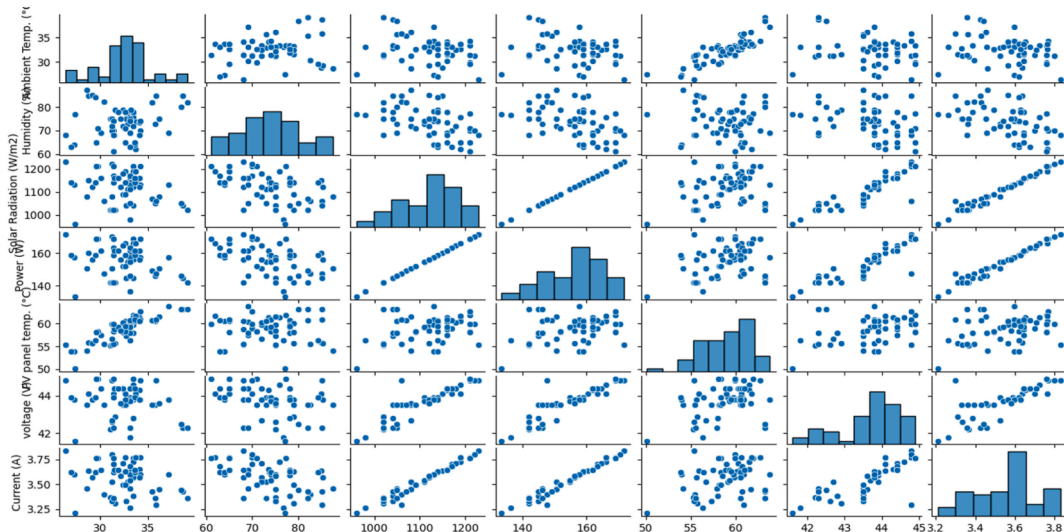


Fig. 4. Correlation pair plot of all the feature vectors.

Table 3
Dataset description.

Data Descriptive Statistics	Ambient Temperature (°C)	Humidity (%)	Solar Radiation (W/m ²)	Power (W)	PV panel temperature (°C)	Voltage (V)	Current (A)
Count	60.0000	60.0000	60.0000	60.0000	60.0000	60.0000	60.0000
Mean	32.4850	73.7566	1118.5000	155.9601	58.8654	43.7146	3.5631
Standard Deviation	2.6436	6.5341	64.67193	9.1187	2.7457	0.8244	0.1510
Minimum	26.4000	61.0000	960.0000	133.6116	50.1349	41.6000	3.2118
25 %	31.2000	69.0000	1067.5000	148.7691	57.1299	43.5000	3.4500
50 %	32.8000	74.0000	1130.0000	157.5816	59.3799	43.8000	3.5977
75 %	33.6000	78.0000	1160.0000	161.8116	60.6674	44.4000	3.6444
Maximum	39.1000	87.5000	1230.0000	171.6816	63.7449	44.9000	3.8321

Table 4
Correlation among the different feature vectors.

	Ambient Temperature (°C)	Humidity (%)	Solar Radiation (W/m ²)	Power (W)	PV panel temperature (°C)	Voltage (V)	Current (A)
Ambient Temperature (°C)	1.0000	0.1709	−0.2113	−0.2113	0.8483	−0.1642	−0.2146
Humidity (%)	0.1709	1.0000	−0.4523	−0.4523	−0.0805	−0.3158	−0.4825
Solar Radiation (W/m²)	−0.2113	−0.4523	1.0000	1.0000	0.3382	0.9064	0.9839
Power (W)	−0.2113	−0.4523	1.0000	1.0000	0.3382	0.9064	0.9839
PV panel temperature (°C)	0.8483	−0.0805	0.3382	0.3382	1.0000	0.3329	0.3263
Voltage (V)	−0.1642	−0.3158	0.9064	0.9064	0.3329	1.0000	0.8169
Current (A)	−0.2146	−0.4825	0.9839	0.9839	0.3263	0.8169	1.0000

values, thereby safeguarding the integrity of the data. This was succeeded by a comprehensive exploratory data analysis (EDA), offering nuanced insights into the intricate interrelations among diverse feature vectors, as visually elucidated through correlation plots. Subsequent to this stage, correlation analysis was conducted to further enhance the dataset's quality, strategically retaining pivotal vectors for modeling purposes while discarding those exhibiting high correlations.

5.3. Time series data of module temperature

The module temperature plays a crucial role in determining the efficiency and performance of photovoltaic (PV) panels. To provide insights into the temporal variation of module temperature over the course of the study, we present a time series analysis of module temperature data collected during the 60-day study period. The variation of module temperature with respect to the day of the study is presented in Fig. 5. The maximum, minimum, and average module temperatures recorded throughout the study period were 63.74 °C, 50.13 °C, and 58.06 °C, respectively.

This figure illustrates the temporal variation of module temperature recorded over the 60-day study period. Each data point represents the module temperature measured on a specific day, providing valuable insights into the dynamics of temperature fluctuations and their potential impact on solar panel performance. Analyzing this temporal variation is essential for understanding its influence on solar panel efficiency and for developing accurate predictive models.

5.4. Stationarity check

To verify the suitability of our dataset for solar power prediction, we conducted a Dicky-Fuller test to assess the stationarity of different feature vectors. Fig. 6 visually presents the stationarity results, ensuring that our data meets the necessary criteria for accurate modeling.

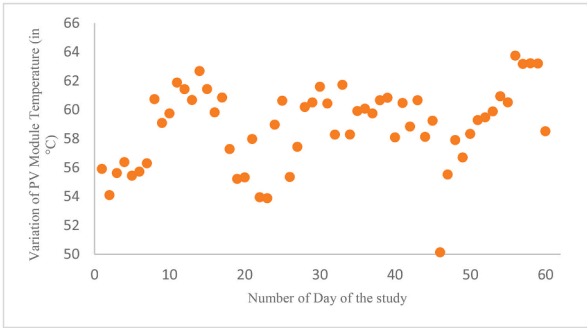


Fig. 5. Time series analysis of module temperature.

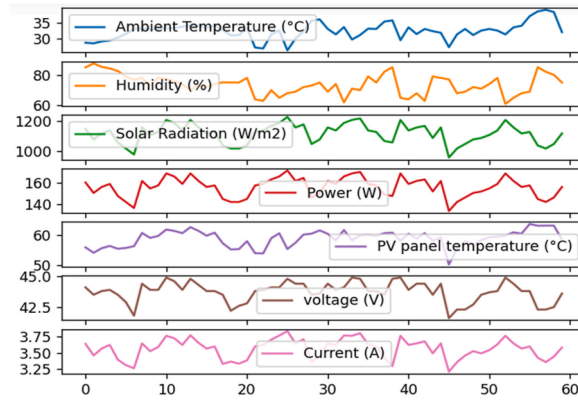


Fig. 6. Representation of different feature vectors.

5.5. Dataset splitting and cross-validation

After confirming the quality of our dataset, a critical step in ensuring reliable model performance, we proceeded to carefully split it into two parts: 75 % for training and 25 % for testing. This division allows the machine learning algorithms to learn patterns from the majority of the data while providing an independent subset for assessing their predictive capabilities.

To enhance the robustness of our model evaluation, we employed a 4-fold cross-validation strategy. In this approach, the dataset is partitioned into four subsets, with the training and testing process repeated four times. During each iteration, three folds are used for training, and one fold is reserved for testing. This iterative process helps to mitigate the impact of variations in the dataset, ensuring that the model's performance is consistently evaluated across different subsets.

The choice of a 75 %/25 % split and 4-fold cross-validation aligns with best practices in machine learning model evaluation. It strikes a balance between providing sufficient data for training to capture underlying patterns and allocating a separate portion for testing to gauge the model's generalization capabilities. This approach enhances the reliability and validity of our findings, contributing to a more robust assessment of the machine learning algorithms applied in our study.

5.6. Algorithm implementation

Various regression algorithms such as multivariate regression, Gaussian regression, support vector regression (SVMR) were implemented. For data cleaning, exploratory data analysis and implementation of various algorithms we have used Python 3.11.1 language and its various libraries like matplotlib and seaborn for exploratory data analysis, Pandas for cleaning and working with dataset and Sklearn library for implementation of various algorithms. The performance of the different algorithms is performed using 4 fold cross validation where 3 folds were used for training and 1 fold for testing.

5.6.1. Support vector machine regression (SVMR)

The application of Support Vector Machine Regression (SVMR) in solar power prediction has yielded remarkable results, as evidenced by the actual and predicted power values for fold 1, fold 2, and fold 3, as illustrated in Fig. 7, Fig. 8, and Fig. 9, respectively. In these figures, a striking alignment between predicted and actual power values becomes evident, signifying the robust predictive capabilities of the SVMR algorithm.

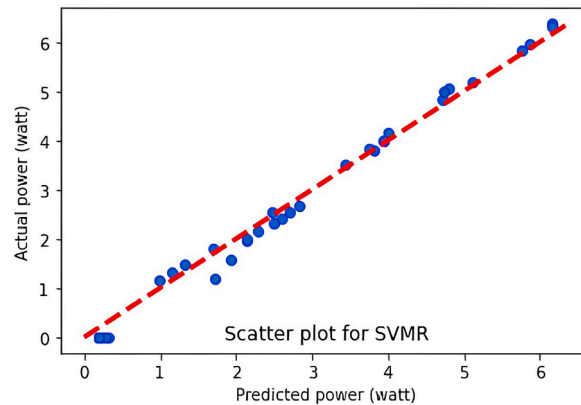


Fig. 7. Scatter plot of actual and predicted power for fold-1 in SVMR.

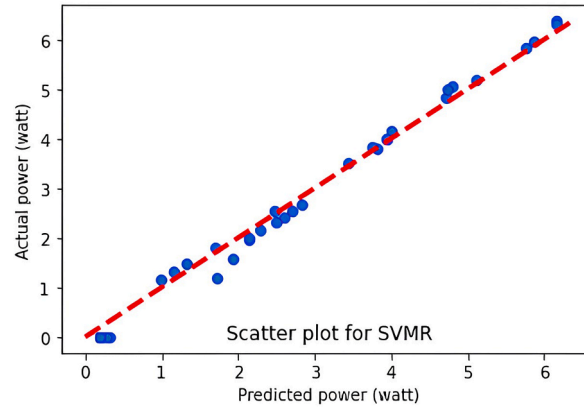


Fig. 8. Scatter plot of actual and predicted power for fold-2 in SVMR.

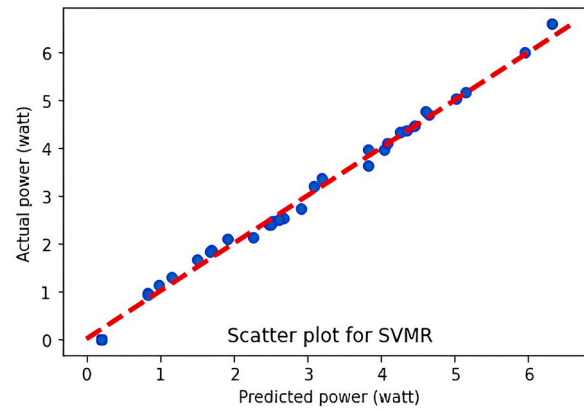


Fig. 9. Scatter plot of actual and predicted power for fold-3 in SVMR.

The performance metrics further underscore the efficacy of SVMR, with an impressive R^2 value of 0.99. This high R^2 value signifies the exceptional degree of correlation between the predicted and actual power values when employing SVMR, surpassing the performance of multivariate regression and Gaussian regression. Additionally, the Mean Absolute Error (MAE) and Mean Squared Error (MSE) in the case of SVMR stand at 0.17 and 0.038, respectively, affirming the accuracy and precision of this regression algorithm. The exceptional performance of SVMR underscores its potential as a highly effective tool for solar power prediction, positioning it as a leading contender among the algorithms assessed in this study.

5.6.2. Gaussian regression

In the context of solar power prediction, Gaussian regression has been evaluated, and the performance results are presented through Fig. 10, Fig. 11, and Fig. 12, showcasing the actual and predicted power values for fold 1, fold 2, and fold 3. When employing Gaussian regression, the coefficient of determination, denoted as R^2 , is calculated to be 0.884. While this R^2 value signifies a notable level of correlation between predicted and actual power values, it is lower in comparison to the R^2 value achieved with SVMR. This implies that Gaussian regression, while demonstrating a reasonable degree of predictive power, falls short of the predictive accuracy achieved with SVMR.

Moreover, the Mean Absolute Error (MAE) and Mean Squared Error (MSE) associated with Gaussian regression are calculated to be 0.63 and 0.49, respectively. These metrics indicate the extent of error in the predictions made by the Gaussian regression model, with MAE measuring the average absolute error and MSE reflecting the average squared error.

5.6.3. Multivariate regression

The performance of the multivariate regression algorithm in the context of solar power prediction has been thoroughly evaluated and is presented through Fig. 13, Fig. 14, and Fig. 15, which offer insights into the algorithm's performance across fold 1, fold 2, and fold 3, respectively. In the case of multivariate regression, the coefficient of determination (R^2) value was computed to be 0.93. This R^2 value signifies a strong correlation between the predicted and actual power values. It highlights the ability of the multivariate regression algorithm to provide accurate predictions of solar power, as evidenced by the substantial R^2 value.

Furthermore, the Mean Absolute Error (MAE) and Mean Squared Error (MSE) associated with multivariate regression have been determined to be 0.39 and 0.28, respectively. These metrics provide valuable insights into the accuracy and precision of the multi-

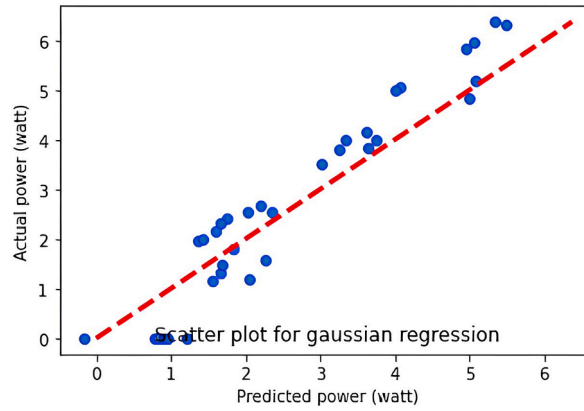


Fig. 10. Scatter plot of actual and predicted power for fold-1 in Gaussian regression.

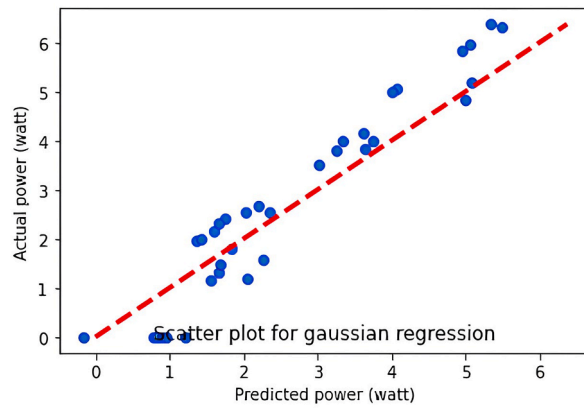


Fig. 11. Scatter plot of actual and predicted power for fold-2 in Gaussian regression.

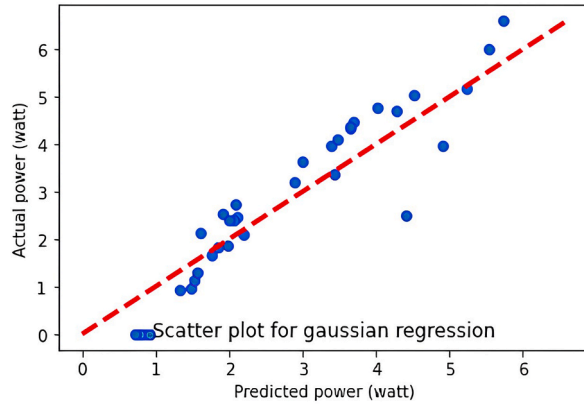


Fig. 12. Scatter plot of actual and predicted power for fold-3 in Gaussian regression.

variate regression model. The MAE represents the average absolute error in the predictions, while the MSE quantifies the average squared error.

The outcomes of this study underscore the critical importance of selecting an appropriate machine-learning model for accurate solar power prediction. Within the context of our dataset, the comparative analysis reveals distinct performances among the applied machine learning models.

5.6.3.1. SVMR and multivariate regression. In our investigation, Support Vector Machine Regression (SVMR) and multivariate regression emerge as highly effective choices, showcasing robust predictive capabilities and accuracy. The high R^2 value, along

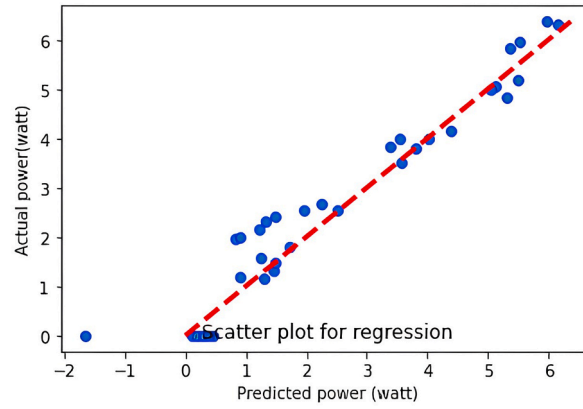


Fig. 13. Scatter plot of actual and predicted power for fold-1 in multi variate regression.

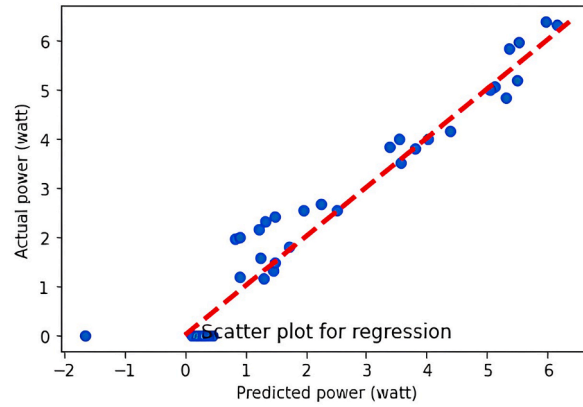


Fig. 14. Scatter plot of actual and predicted power for fold-2 in multi variate regression.

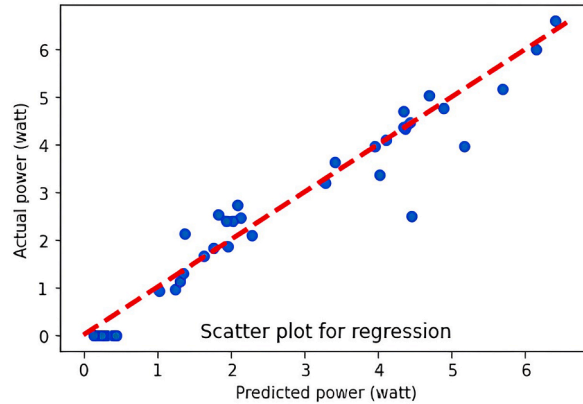


Fig. 15. Scatter plot of actual and predicted power for fold-3 in multi variate regression.

with low Mean Absolute Error (MAE) and Mean Squared Error (MSE), attests to the reliability of these models in capturing and predicting solar power output. Their strong performance positions them as valuable tools for precise solar power predictions.

5.6.3.2. Gaussian regression. Conversely, the Gaussian regression model demonstrates a comparatively lower predictive capacity, evident in its lower R^2 value and higher error metrics. While still providing a reasonable level of predictive power, the model falls short of the accuracy achieved by SVMR and multivariate regression. The findings underscore the importance of careful model selection, with Gaussian regression showing limitations in capturing the complexities of the solar power prediction task in this particular dataset.

5.7. Practical implications

The effectiveness and accuracy of the proposed machine learning models, particularly SVMR and multivariate regression, hold substantial practical implications. These models offer reliable methods for predicting solar PV system output power, a crucial factor in optimizing the performance and economic viability of solar power plants. The study's findings provide valuable insights for decision-makers in choosing appropriate models for solar power prediction tasks.

5.8. Recommendations

In summary, the comparative analysis indicates that SVMR outperforms both multivariate regression and Gaussian regression in terms of R^2 , MAE, and MSE. However, the choice of the most suitable model should be tailored to the specific characteristics of the dataset and the desired level of predictive accuracy. Researchers and practitioners are encouraged to carefully consider these factors when implementing machine learning models for solar power prediction tasks.

6. Comparative analysis

6.1. Methodological comparison

In comparison to Ahmed et al. (2021), who focused on classifying PV panels using an isolated convolution neural model (ICNM) based on infrared thermographic images, our research took a different approach by employing Support Vector Machine Regression (SVMR), multivariate regression, and Gaussian regression for solar power prediction. Borujeni et al. (2022) utilized ML algorithms like SVM and Gaussian process regression (GPR) for predicting PV power based on various input parameters, while Prema et al. (2022) conducted a comprehensive review of short-term wind and solar power forecast models. Our study stands out for its methodological diversity, exploring the effectiveness of different machine learning models on a diverse dataset under fluctuating environmental conditions.

6.2. Performance metrics

While Ahmed et al. (2021) focused on accuracy in classifying PV panels based on health and defects, Borujeni et al. (2022) emphasized metrics such as RMSE, MAE, and R^2 , with Matern 5/2 GPR providing the highest performance. In contrast, our research demonstrated the superiority of SVMR over multivariate regression and Gaussian regression based on R^2 , MAE, and MSE metrics. Each study contributed valuable insights into model performance, with our research showcasing the effectiveness of SVMR in the context of solar power prediction.

6.3. Dataset characteristics

Ahmed et al. (2021) utilized a dataset focused on infrared thermographic images for PV panel classification, while Borujeni et al. (2022) considered input parameters like solar PV panel temperature, ambient temperature, solar flux, time of day, and relative humidity for predicting PV power. In our study, a comprehensive dataset including ambient temperature, humidity, solar radiation, panel temperature, voltage, and current was employed for solar power prediction. This diversity in dataset characteristics allows for a nuanced comparison and highlights the versatility of machine learning models.

6.4. Innovations and contributions

Ahmed et al. (2021) contributed by introducing the isolated convolution neural model (ICNM) for accurate PV panel classification, while Borujeni et al. (2022) introduced an intelligent prediction-based controller using ML algorithms, emphasizing the high reliability and accuracy of the developed Matern 5/2 GPR model. In contrast, our research contributed through the application of SVMR, multivariate regression, and Gaussian regression for solar power prediction, considering a diverse dataset with a range of environmental variables. Each study addressed specific challenges and added unique insights to the broader field of renewable energy research.

6.5. Implications and recommendations

Ahmed et al. (2021) suggested implications for savings in PV investment by accurately classifying panels based on health and defects. Borujeni et al. (2022) recommended the use of ML models as rapid tools for predicting the suitable performance of solar PV panels, with further studies recommended for additional parameters. In our research, we emphasized the practical implications for optimizing the performance and economic viability of solar power plants through accurate solar power predictions. Each study provided valuable recommendations for the application and future exploration of machine learning models in solar energy scenarios.

The current study presents a robust forecasting model for Solar PV panels, leveraging variations in environmental parameters to accurately predict output power. By focusing on real-time environmental influences, the model offers valuable insights for optimizing PV system performance in the short term. However, it is essential to acknowledge the long-term implications of photovoltaic module degradation, which can impact efficiency over the 20–25 year lifecycle. While the model developed in this study demonstrates accuracy within current environmental conditions, future research directions should explore advanced techniques such as deep learning or image-based learning. These advanced methodologies hold promise for incorporating historical performance data and degradation rates, enabling more accurate predictions over the entire lifespan of PV modules. By integrating degradation modeling into forecasting approaches, future studies can enhance the accuracy and reliability of predictions, facilitating optimal system maintenance and maximizing energy yield over the module's 25-year lifespan.

6.6. Techno-economic and environmental assessment

This research presents a comprehensive analysis of a solar power system's performance, incorporating experimental data collection, machine learning modeling, and techno-economic and environmental assessment. The study emphasizes the critical role of accurate prediction models in optimizing solar power generation efficiency, with support vector machine regression emerging as the most effective algorithm. The techno-economic assessment underscores the financial viability of the proposed solar power system, while the environmental assessment highlights its potential to mitigate climate change and reduce environmental pollution. Overall, the research contributes valuable insights into the design, implementation, and impact of solar power systems, offering practical guidance for maximizing their economic and environmental benefits.

7. Practical applications and future research prospects

The application of machine learning algorithms for solar power prediction holds significant potential in various real-world scenarios. One practical application includes the optimization of solar power plant performance, where accurate forecasting models can help operators better manage energy generation and distribution. Moreover, these models can aid in grid integration and energy storage management, contributing to the stability and efficiency of renewable energy systems. Beyond operational optimization, our study opens avenues for future research in several areas. One direction involves the refinement and enhancement of prediction models to accommodate dynamic environmental conditions and emerging technologies. Furthermore, exploring the integration of IoT sensors and satellite imagery for real-time data collection and analysis could further improve the accuracy and reliability of solar power forecasts. Additionally, investigating the economic feasibility and scalability of implementing these models across different geographical regions and solar installations presents another promising avenue for research. By addressing these practical applications and future research prospects, our study seeks to not only contribute to the advancement of solar power forecasting but also inspire further innovation in renewable energy management and sustainability efforts.

8. Conclusions

This study underscores the paramount importance of accurate prediction models in maximizing the potential of solar power systems amidst the inherent variability in photovoltaic (PV) module performance. Through meticulous research and analysis, the findings reveal that support vector machine regression (SVMR) emerges as the top-performing algorithm. With an exceptional R^2 value of 0.99 and minimal errors, SVMR demonstrates superior predictive capabilities for output power. Additionally, multivariate regression yields commendable results, while Gaussian regression provides valuable insights despite slightly lower performance.

The implications of these findings extend beyond the geographical confines of southern India, offering valuable insights for solar companies globally. By leveraging real-world data from a detailed site survey, the study enhances the accuracy of prediction models, particularly through the innovative use of SVMR. In summary, this research not only provides practical guidance for optimizing solar power plant performance but also highlights the efficacy of machine learning in solar energy generation. The potential for widespread adoption of these models further solidifies the study's significance in advancing sustainable energy practices.

CRedit authorship contribution statement

Abhishek Kumar Tripathi: Writing – review & editing, Writing – original draft, Resources. **Mangalpady Aruna:** Writing – review & editing, Writing – original draft, Supervision. **P.V. Elumalai:** Formal analysis, Conceptualization. **Krishnasamy Karthik:** Conceptualization. **Sher Afghan Khan:** Formal analysis. **Mohammad Asif:** Formal analysis. **Koppula Srinivas Rao:** Formal analysis.

Declaration of competing interest

None.

Data availability

No data was used for the research described in the article.

Acknowledgement

The financial support from the Researchers Supporting Project (RSP2024R42), King Saud University, Riyadh, Saudi Arabia is appreciated. The authors express their sincere gratitude towards the Department of Mining Engineering and the management of Aditya Engineering College, Surampalem, India for providing the research facilities and necessary support during entire phase of this research work.

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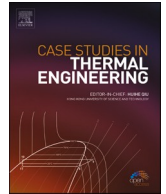
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Corrigendum to “Advancing solar PV panel power prediction: A comparative machine learning approach in fluctuating environmental conditions” [Case Stud. Therm. Eng. 59 (July 2024) 104459]

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The authors would like to apologise for any inconvenience caused.

DOI of original article: <https://doi.org/10.1016/j.csité.2024.104459>.

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<https://doi.org/10.1016/j.csité.2024.104559>

Available online 16 May 2024

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